

inclinations, face another formidable obstacle: front-running costs that come from regularly rolling over their commodity futures positions. They occur when others anticipate and trade in front of the futures rollover dates, then take profits afterward.

Yiqun Mou (2011) estimates that front-running costs were 3.6% annually from January 2000 through March 2010. J.P. Morgan Commodities Research reported in 2009 that roll returns have put a drag of 3% to 4% per year on commodity index returns since 1991. These hidden costs can quickly take the wind out of the sails of the buyers of passive commodities index futures.

Rising correlations are another problem associated with commodities. According to Tang and Xiong (2012), the average one-year rolling correlation among indexed commodities was at a stable level below 0.10 throughout the 1990s and 2000s. By 2009, it had climbed to 0.50. Before 2008, the correlation between the GSCI and the S&P 500 was generally in a band between -0.20 and 0.10 . Since then, it has shot up to over 0.50 and has generally remained there.¹³

Furthermore, during both the 1929 stock market crash and the 2008 financial crisis, the correlation between equities and commodities shot up to over 80%. Commodities diversification was lacking when it was needed the most. According to Lombardi and Ravazzolo (2013) of the Bank for International Settlements, the popular view that commodities should be included in one's portfolio as a hedging device is no longer valid.¹⁴

The decrease in roll return and increase in return correlation now render the mean-variance diversification benefit of a passive allocation to commodity futures insignificant. A comprehensive study by Daskalaki and Skiadopoulos (2011) called "Should Investors Include Commodities in Their Portfolios After All? New Evidence," revealed that the introduction of commodity instruments in a traditional stock/bond portfolio is no longer beneficial for a utility-maximizing investor. Blitz and de Groot (2004) also found that commodities deserve little or no role in a stock/bond portfolio, although a case could be made for including momentum, carry, and low-volatility commodity market risk factors.

The first year that investors could actually include a passive commodity index in their portfolios was 1991. [Table 5.1](#) shows the performance of the DJ-UBSCI, compared with the performance of the S&P 500, Morgan Stanley Capital International Europe, Australasia, and Far East (MSCI